Krishna Defence System / Garuda AI

TECHNICAL DESIGN & ARCHITECTURE DOCUMENT — HACK FOR SOCIAL CAUSE 2026

Theme: AI, GIS & Emerging Technologies | Proactive Cyber Threat Forecasting & Decision Support

1. Executive Summary: World-Model Predictive Cyber Defence

Learn the evolving network state, simulate multiple future states, explain the risk trajectory, and route evidence into controlled defensive action.

Krishna Defence System is the complete platform. Garuda AI is its forecasting core. The separate Krishna Investigate lane handles unknown / unsupported evidence so the platform name and the triage function are not confused.

2. End-to-End System Pipeline

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PCAP/Flow Telemetry → Feature Pipeline (34 Metrics) → Graph Network State S_t
→ Spatial GNN (GraphSAGE) → Garuda AI Temporal LSTM Dynamics → Autoregressive Rollout (+10s, +20s, +30s, +40s)
→ MITRE Stage & XAI Attribution → Controlled Defence Router (Arjuna / Krishna / Sudarshana)
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3. Subsystem Responsibilities & Safety Boundaries

Subsystem	Responsibility	Output / Safety Boundary
Telemetry + Parser	Ingests PCAP/flow telemetry; normalizes timestamps into schema.	Unsupported formats or missing evidence fail explicitly.
State / Graph Builder	Creates 10-second graph snapshots and ordered histories.	Sequence gaps break histories; future windows never leak.
Garuda Core	GraphSAGE encodes relations; LSTM learns transition dynamics P(S_t+1 | S_t).	State forecasting remains separate from risk/stage claims.
Forward Simulation	Autoregressively rolls state forward (+10s, +20s, +30s, +40s).	Future state + uncertainty + risk; not a guaranteed compromise.
Interpretability / Stage	Ranks local feature drivers; MITRE stage only with supervised support.	Unresolved evidence is preferred over an invented stage.
Defence + Audit	Arjuna = known response; Krishna = triage; Sudarshana = containment.	Human/policy gates, expiry, revoke/kill-switch prevent blind blocking.

4. Feasibility, Performance & Societal Impact

• **Offline-First Architecture:** Zero cloud or external API dependencies; executes completely on standard local CPU hardware.
• **Ultra-Low Latency:** ~2.9 ms mean forecast latency (332–349 calls/sec) with ~45 MiB max RSS memory footprint.
• **Fail-Closed Safety:** Novel or out-of-distribution telemetry defaults to advisory shadow mode rather than triggering false-positive containment.
• **Verified Results:** ~27.5% lower state MSE vs persistence (CICAPT-IIoT2024); ~98.7% recall with ~0.29% FPR (UNSW-NB15).
• **Societal Impact:** Democratizes high-grade cyber defence for small utilities, municipal governance, public schools, and healthcare infrastructure.